

Vision-based UAV Detection under Adverse Weather Conditions

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This researcher investigates vision-based UAV detection under challenging environmental conditions like rain and image distortions, including motion blur and noise, negatively impact detection performance . the YOLOv5m model's mAP50 score drops from 95.6% on clean data to 66.8% in drizzle and further down to 47.9% in torrential rain,. Similar to rain, higher levels of motion blur cause greater degradation of mAP. In testing, the mAP50 for YOLOv5m plummeted from 95.6% to 21.9% under high motion blur conditions,.The researchers developed a Complex-Background Dataset (CBD) featuring 10,000 images with diverse urban and natural environments, including flying birds to increase difficulty. Also created specialized test sets by artificially adding rain (drizzle to torrential), motion blur, and Gaussian noise to clean images to simulate real-world environmental challenges—. They tested five prominent deep learning algorithms -YOLOv5, YOLOv8, YOLO-NAS, Faster-RCNN, and RetinaNet- to see how their accuracy (mAP) and speed were affected by these conditions,. They proposed a training strategy using an Augmented Complex-Background Dataset (ACBD), which mixed clean images with 33% noisy, 33% blurred, and 34% rainy samples to "enhance" the models.YOLOv5 (specifically the YOLOv5m version) was the top-performing object detection model across the majority of the study's tests, followed by YOLOv8m, evaluations using the Complex-Background Dataset (CBD) without distortions, YOLOv5m achieved YOLOv5m: 95.6% mAP50, YOLOv8m: 91.8% mAP50 ,YOLO-NASs: 88.17% mAP50 , RetinaNet: 84.39% mAP50 ,Faster-RCNN: 82.41% mAP50 researchers also tested the Enhanced versions of these models by training them on an augmented dataset (ACBD) containing the distorted images. In these tests, Enhanced YOLOv5m remained the best performer, significantly improving its scores across all adverse categories. In torrential rain, the Enhanced YOLOv5m improved its mAP50 from 47.9% to 83.3%.Beyond accuracy, YOLOv5 was also the most efficient model in terms of speed, in the inference time YOLOv5 was the fastest, taking only 7.1 ms per image, compared to YOLOv8 (9.5 ms), Faster-RCNN (73.38 ms), and RetinaNet (153.18 ms), and in the training speed: YOLOv5 achieved the highest training rate at 4.54 iterations per second.